Geophysical Journal (1988) **95**, 69–78

An early Proterozoic VGP from an oriented drillcore into the Precambrian basement of Southern Alberta

Özden Özdemir, David J. Dunlop, Brian Reid and Hironobu Hyodo

Geophysics Laboratory, Department of Physics, Erindale College, University of Toronto, Toronto, Ontario, M5S 1A7 *Canada*

Accepted 1988 March 23. Received 1988 March 21; in original form 1987 July 29.

SUMMARY

We report paleomagnetic results for 25 samples from a fully-oriented PanCanadian Petroleum drillcore which penetrated the Precambrian basement of southern Alberta at a depth of 2550 m. Samples were taken at approximately 1 m intervals over a 7 m section of core. Intensities of natural remanent magnetization (NRM) were 0.5–14 A m^{-1} over the section, which consists mainly of granodiorite, with one short granite interval. X-ray and Curie temperature data indicated magnetite as the only important NRM carrier. Most of the NRM was an intense, steeply-inclined drilling remanence, with coercivities <20 mT and distributed low-to-intermediate unblocking temperatures. After magnetic cleaning, a weak but stable primary NRM with high coercivities and high unblocking temperatures was isolated, with $D = 157°$, $I = +52°$ ($k = 118$, $\alpha_{95} = 3°$, $N = 20$). The corresponding virtual geomagnetic pole (VGP) is 4°S, 93°W ($dp = 3°$, $dm = 4°$), which falls on the North American apparent polar wander path around 1850 Ma. The basement of this part of Alberta is probably an extension of the Churchill Structural Province of the Laurentian Shield.

Key words: Churchill Province, oriented drillcore, palaeomagnetism, Proterozoic basement

1 INTRODUCTION

Because of the importance of testing the Vine & Matthews' (1963) hypothesis, there has been lively interest in the paleomagnetism of igneous basement cores drilled in the Deep Sea Drilling Project (DSDP). Since DSDP cores are unoriented in azimuth, only the inclination of natural remanent magnetization (NRM) can be determined. There have been comparatively few paleomagnetic studies of continental sedimentary sections and igneous basement sampled by boreholes (Van der Voo & Watts 1978; Kodama 1984; Evans & Maillol 1986). In most cases, azimuthal orientation of the cores and NRM declinations were determined indirectly, using sedimentary fabric or secondary magnetization directions.

The drillcore studied in this paper (PCP Travers 1-31-13-20W4) was fully oriented by the Christensen–Hügel method. It was drilled by the PanCanadian Petroleum Company through a 2531 m sedimentary section in southern Alberta (50.1°N, 112.7°W). Underlying the basal Cambrian sandstone and a weathered interval at the unconformity was fresh unweathered granodiorite, which was cored from 2550 to 2563 m. A 7 m section from the base of this igneous core is studied here. Since the drillhole is located several hundred km east of the Foothills Thrust Belt, tectonic rotation of the granodiorite pluton is unlikely and we have made no tilt corrections to the data that follow.

This part of the basement of Alberta is thought to belong to the Churchill Structural Province of the Canadian Precambrian Shield (Fig. 1). Rocks of the Churchill Province tend to have metamorphic ages between 1900 and 1700 Ma. The age determined paleomagnetically for the present core falls in this interval.

2 SAMPLING AND MEASUREMENT PROCEDURES

Twenty-five transverse cores 2.5 cm in diameter were drilled at eight equally spaced positions along the main core (10 cm diameter). Four specimens for paleomagnetic study each 2.3 cm long were cut from each transverse core. XRF and petrographic analyses were carried out on material from each level (Boyle 1985). Except for a short pink granite interval around 2560 m, the samples are granodiorites from a large sub-jacent pluton. There is scattered-to-extensive alteration of mafic minerals to chlorite and occasionally magnetite. Chlorite/biotite foliation is weakly developed. The metamorphic grade is greenschist, implying regional reheating to no more than 450–500 °C.

The core was oriented by the Christensen–Hügel method (Rowley *et al.* 1981). A multishot survey instrument containing a magnetic compass was located within a non-magnetic drill collar above the core barrel and was photographed every 1–2 min. Eight shots (orientations) were taken at 1 m intervals from 2556 to 2563 m, plus two extra shots at 2559.6 and 2563.2 m, for a total of 10 shots over the 7 m interval studied. The survey instrument was aligned with the inner core barrel by an extension rod.

Downloaded from https://academic.oup.com/gji/article/95/1/69/602275 by Curtin University Library user on 26 May 2026

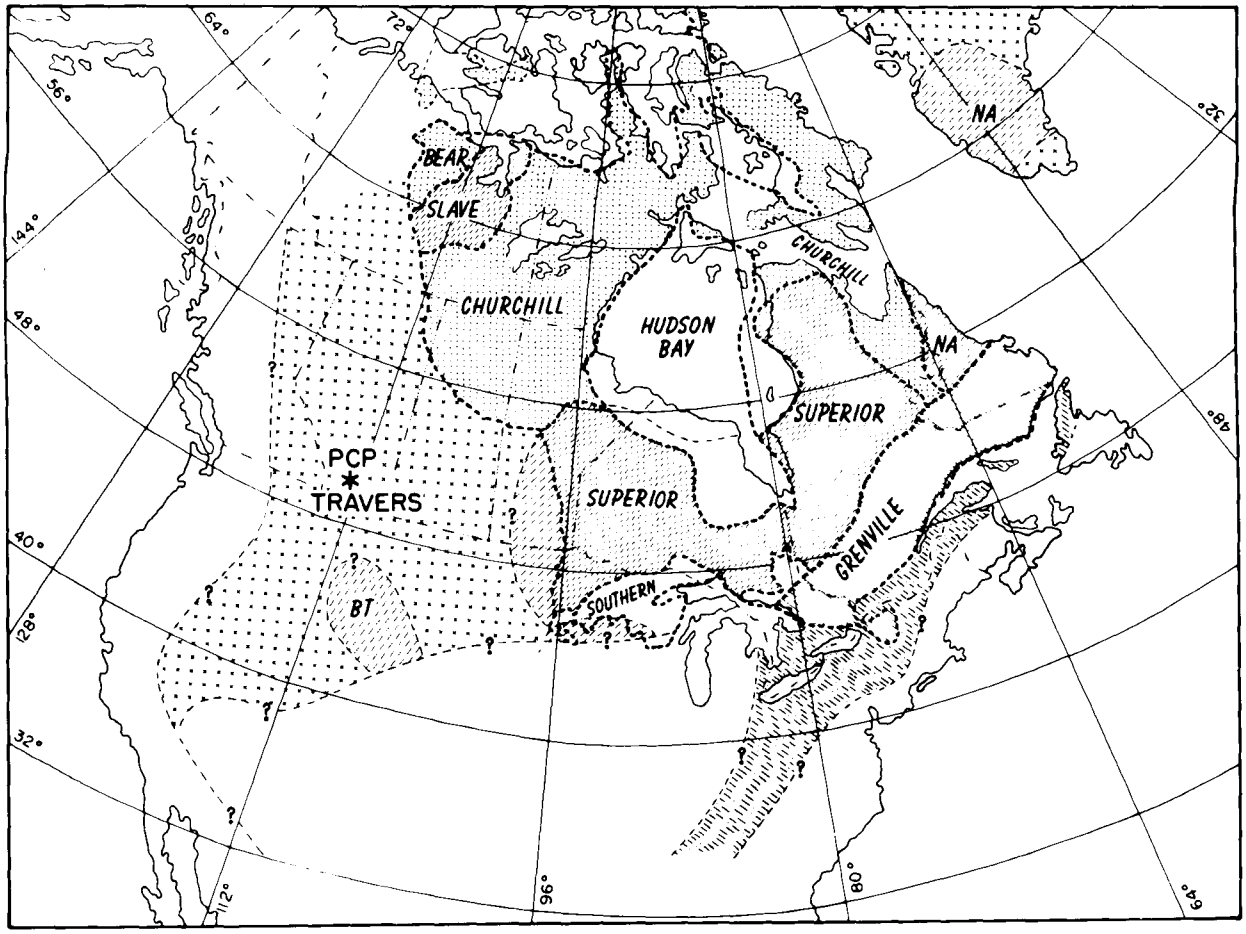


Figure 1. A sketch map of the structural provinces of the Canadian Precambrian Shield and their possible extensions under Phanerozoic cover (adapted from Irving & McGlynn 1981, fig. 23.3). Exposed parts of the shield and province boundaries are outlined by heavy dashed lines. NA, BT denote the North Atlantic and Beartooth (Wyoming) cratons respectively. Possible extensions of the Churchill Province and other middle Proterozoic reactivated terrains under younger cover are indicated by a pattern of *x*s. The PCP Travers well lies in such a region.

Grooves were cut continuously into the drillcore as it entered the barrel by a scribing shoe. An orienting lug with a fixed, known orientation relative to the grooves appears in each photograph of the compass.

Frictional torque between the inner and outer core barrels causes spiralling of the grooves down the core (Halgedahl & Jarrard 1988). In our core, groove orientation changed about $10° m^{-1}$ over continuous sections. There were two jumps in the reference groove orientation log, between 2557 and 2558 m and between 2561 and 2562 m, corresponding to breaks in the core. Interpolations to determine azimuth at the levels sampled paleomagnetically were made along unbroken sections over which the reference groove orientation varied smoothly. Declinations of characteristic NRM, reported in detail in Section 6, varied randomly from 142° to 173° (mean 157°, standard deviation 9.3°) and were uncorrelated with variations in multishot orientations.

Multishot orientation is generally accepted as the industry standard, and is capable under ideal circumstances of an accuracy of ±5°, although ±10°–15° is more typical (Bleakly *et al.* 1985). Nelson *et al.* (1987) have recently given a careful analysis of the errors involved in multishot orientation. Error in the coring and surveying procedure is estimated to be ±5°; reading orientations in the laboratory with a mechanical goniometer has an inherent instrumental error of ±3°, augmented by ±2°–4° of reproducibility error in replicate determinations, depending on whether a single analyst or different analysts made the readings. Overall estimated error is ±11°.

Since orienting errors affect azimuth only, we would expect dispersion in NRM declination D to exceed dispersion in the inclination I if orienting errors are substantial. The inclination of the characteristic NRM in our study varied randomly from 45° to 62° (mean 52°, standard deviation 4.9°). The dispersion in I is about one-half the 9.3° dispersion in D, suggesting that random orienting errors account for about 4°–5° of the dispersion in NRM declinations.

NRM was measured with a Schonstedt spinner magnetometer. One specimen from each sample was demagnetized by alternating fields (AF) in 20 steps to 80 mT and one specimen was thermally demagnetized in 12 steps to 675 °C. Schonstedt demagnetizers were used in both cases. Initial weak-field volume susceptibility was measured using a Bartington bridge.

X-ray analysis was carried out using a Guinier de Wolff camera with Fe–Kα radiation. High-field thermomagnetic

Downloaded from https://academic.oup.com/gji/article/95/1/69/602275 by Curtin University Library user on 26 May 2026

curves were measured with an automatically recording vertical Curie balance. The hysteresis parameters saturation magnetization, M_s, saturation remanence, M_{rs}, and bulk coercive force, H_c, were determined at room temperature with a vibrating-sample magnetometer and a 600 mT electromagnet.

3 ROCK MAGNETIC AND X-RAY STUDIES

X-ray and thermomagnetic analyses were made on magnetic separates. The strongest X-ray peaks were the (220), (311), (400), (333, 511) and (440) lines of magnetite. Cell-edge parameters of 8.405 and 8.416 Å, for samples from the 2555.8 and 2562.7 m levels respectively, are close to the ASTM value of 8.397 Å for pure magnetite.

A typical reversible high-field thermomagnetic curve for one of the magnetic separates is shown in Fig. 2. Curie temperatures of 575, 581 and 581 °C were found for samples from the 2555.8, 2560.0 and 2562.7 m levels, respectively. All are close to the magnetite Curie temperature of 580 °C.

Complete hysteresis curves were determined for three sub-samples. Values of M_s and M_{rs}/M_s for these and 20 additional sub-samples are given in Table 1. Saturation magnetization ranged from 0.167 to 1.524 A m^2 kg^{-1}, indicating 0.2–1.5 per cent by weight magnetite. There was no systematic variation in magnetite content with depth, except that the pink granite horizon had slightly less magnetite than the granodiorites.

Both M_{rs}/M_s values (0.015–0.034) and H_c values (1.7, 1.9 and 2.5 mT for samples 3, 8 and 19, respectively) indicate magnetite grains with multidomain (MD) structure. Strain-free crystals of magnetite in the 1–10 μm size range have similar M_{rs}/M_s and H_c values (Heider *et al.* 1987).

The acquisition of isothermal remanence (IRM) as a function of applied field was measured for 12 whole-rock specimens, using a small electromagnet and the spinner magnetometer. NRM and susceptibility values for the same specimens appear in Table 1. Typical IRM-H curves are shown in Fig. 3. The IRM saturated around 150 mT. M_{rs}

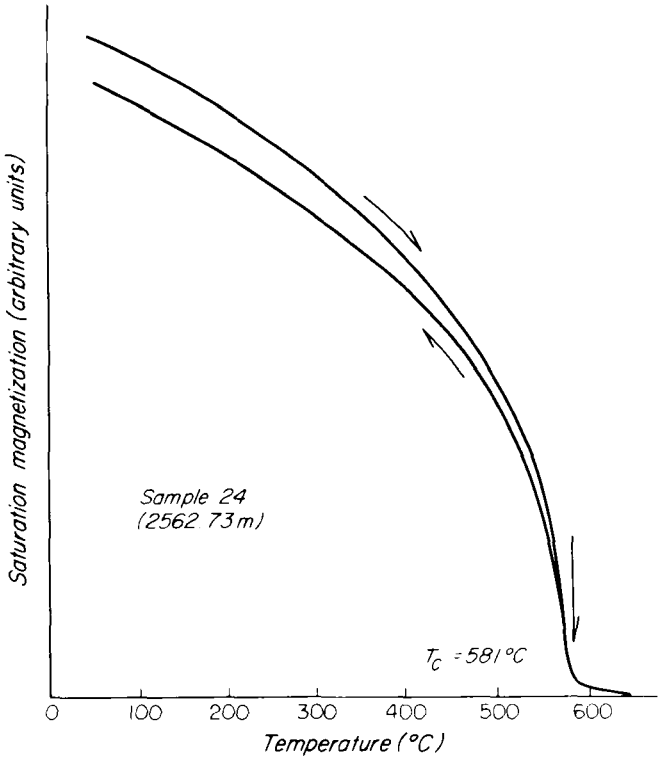


Figure 2. A typical reversible high-field thermomagnetic curve showing the Curie temperature of pure magnetite.

Table 1. Magnetic properties of the samples

No.	Depth (m)	I(°) soft	I(°) hard	*NRM* (A m^{-1})	χ_0 (10^{-2} SIu)	Q_n	*MDF* (mT)	M_s (A m^2 kg^{-1})	M_{rs}/M_s
1.1	2555.8	77	47.5	1.57	1.34	2.4	2.30	0.69	0.024
2.4	2555.9	86	59	6.63	2.38	5.8	1.93	0.83	0.026
3.2	2555.9	70	55	1.79	1.92	2.0	2.30	0.57	0.021
4.2	2556.8	80	55	1.72	2.18	1.6	2.47	1.01	0.023
5.3	2556.9	75	48.5	1.34	1.66	1.7	2.57	0.64	0.028
6.2	2556.9	84	48.5	0.85	1.18	1.5	2.89	1.37	0.023
7.2	2557.8	66	52	1.44	1.36	2.2	2.73	0.72	0.020
8.1	2558.0	86	62	1.19	1.06	2.4	2.96	0.47	0.029
9.2	2558.1	74	50	0.81	0.90	1.9	2.73	1.39	0.016
10.2	2558.2	80	54	2.65	1.84	3.0	2.31	1.39	0.021
11.2	2558.7	76	60	2.59	1.69	3.2	2.27		
12.2	2558.8	84	54	1.32	1.38	2.0	2.91	1.00	0.027
13.2	2558.8	84	54	1.54	1.03	3.1	2.53	0.25	0.024
14.2	2560.0	88		0.58	0.96	1.3	5.17	0.57	0.015
15.1	2560.1	78		0.24	0.43	1.1	6.02	0.68	0.018
16.2	2560.9	72	51	1.50	1.27	2.5	2.33	0.42	0.026
17.2	2561.0	86	57	2.54	2.01	2.6	2.12	1.44	0.021
18.2	2561.0	84	48	1.17	1.12	2.2	2.53	0.80	0.024
19.2	2562.0	84	47	2.25	1.88	2.5	2.57	0.17	0.033
20.3	2562.0	77	46	1.76	1.88	2.0	3.15	0.41	0.034
21.1	2562.1	63	48	2.37	1.72	2.9	2.94	0.93	0.030
22.2	2562.2	77	54	1.71	1.84	1.9	3.22	0.81	0.023
23.4	2562.3	78	41	3.76	2.98	2.6	3.25	1.53	0.030
24.2	2562.7	80	49	2.98	1.75	3.6	2.46		
25.2	2562.8	88	45	3.84	1.54	5.2	2.42	0.54	0.026

Downloaded from https://academic.oup.com/gji/article/95/1/69/602275 by Curtin University Library user on 26 May 2026

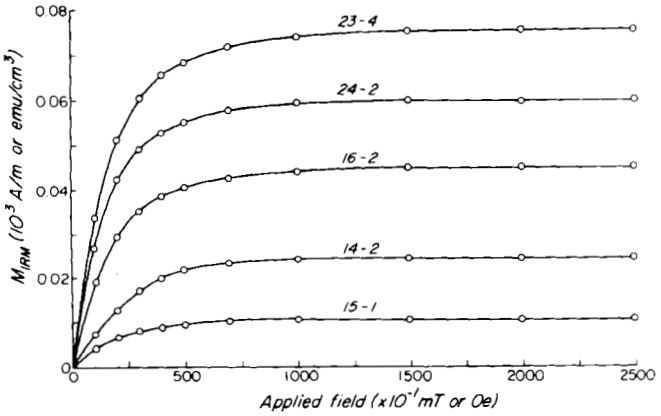


Figure 3. Typical IRM induction curves for samples with a range of magnetite contents.

values ranged from 0.01 to 0.095 kA m^{-1}. Using M_s data from Table 1, M_{rs}/M_s is in the same range determined from hysteresis. These saturation fields and remanence intensities are typical of MD magnetite.

4 NRM AND SUSCEPTIBILITY

The samples were strongly magnetized. Intensities of NRM ranged from 0.2 to 14 A m^{-1}. Specimens from the edges of the main drillcore (specimens 1 and 4) usually had the highest NRM intensities (Fig. 4). Specimens 2 and 3 from

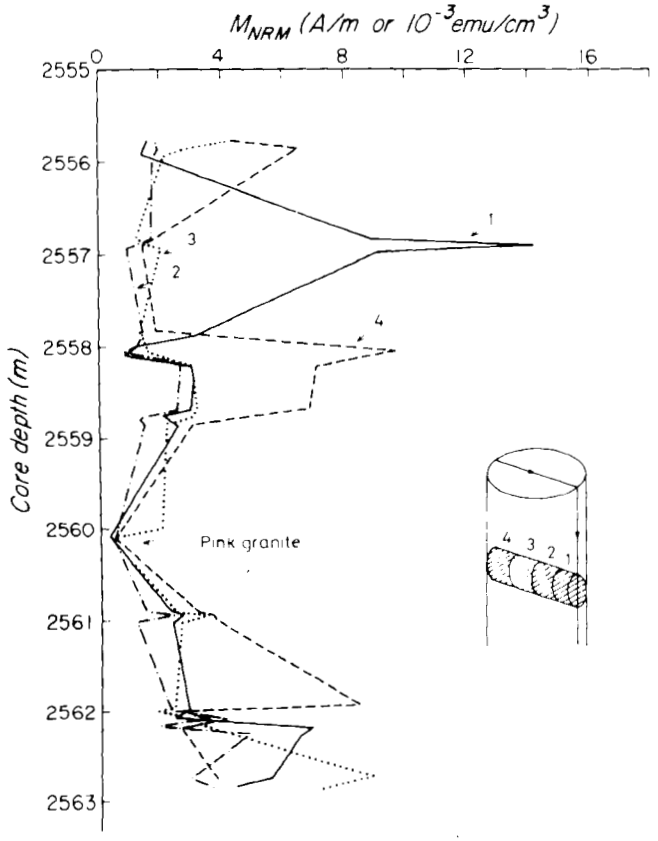


Figure 4. Depth variation of NRM intensity for specimens from the centre and the edges of the main drillcore (see inset).

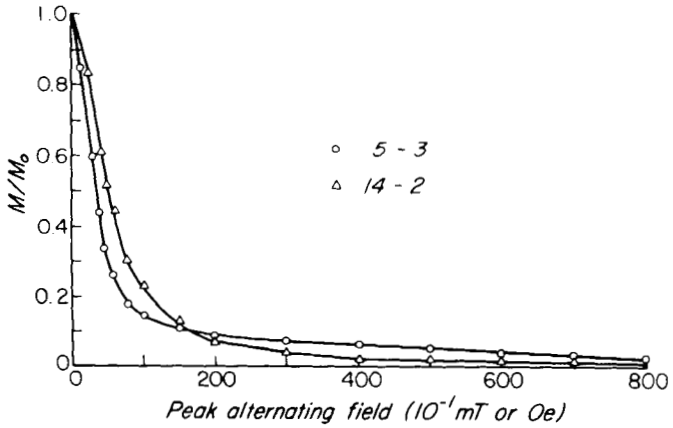


Figure 5. Typical decay of NRM intensity during stepwise AF demagnetization for granodiorites (5.3) and granites (14.2).

the centre of the core had intensities between 1 and 3 A m^{-1}, except for pink granite samples 14 and 15, whose NRMs were in the range 0.2–0.6 A m^{-1}. The central specimens were preferred for paleomagnetic work since their underlying hard stable remanence formed a larger fraction of the bulk NRM.

Values of initial volume susceptibility χ_0 were mainly in the range $1–2 \times 10^{-2}$ ($0.8–1.6 \times 10^{-3}$ emu cm^{-3} Oe^{-1} in cgs) (Table 1). The variation of susceptibility with depth was similar to the depth variation of NRM intensity, resulting in fairly constant values of the Königsberger ratio, $Q_n = M_{nrm}/\chi_0 H$ (Table 1). $H = 50$ A m^{-1} (0.6 Oe) was assumed in calculating Q_n. For the granodiorite samples, Q_n averaged 2.65; for the granites, Q_n was about 1.2. Although low, these values are larger than the values $Q_n < 0.5$ expected for coarse MD magnetite (Stacey 1967).

All 25 samples had soft exponential-like AF decay curves of NRM (Fig. 5), characteristic of MD material (Dunlop

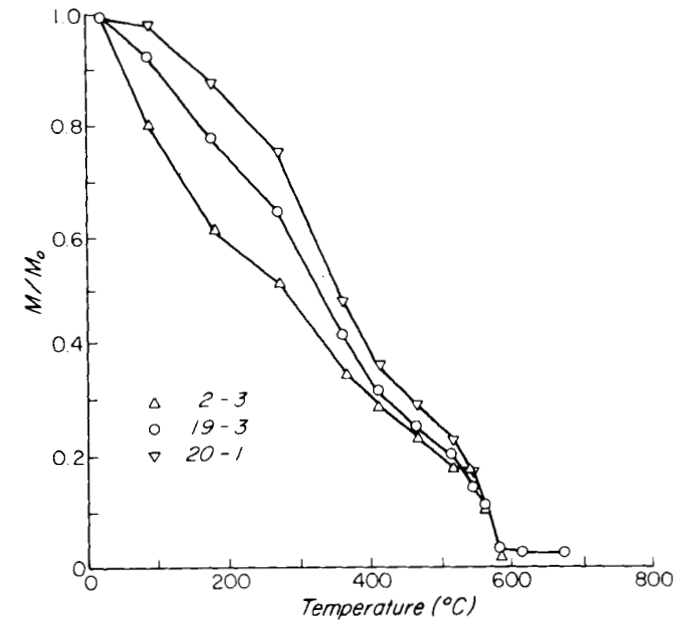


Figure 6. Soft, intermediate, and hard NRM decay curves measured in the course of stepwise thermal demagnetization. All decay curves have bimodal distributions of unblocking temperatures.

Downloaded from https://academic.oup.com/gji/article/95/1/69/602275 by Curtin University Library user on 26 May 2026

1983a). Median destructive fields (MDFs) were 2–3 mT, except for the granite samples (2560 m), whose MDFs were 5–6 mT.

Thermal decay curves of NRM were more variable but generally soft (Fig. 6). The unblocking temperature spectrum is bimodal, with a principal peak below 400 °C and a second, smaller peak just below the magnetite Curie point. These peaks are associated with two distinct NRM vectors, to be discussed in the following sections. Hematite carries little or none of the NRM.

5 DRILLING-INDUCED REMANENCE

Upon AF or thermal demagnetization, the NRM of all samples was found to be bivectorial. Between 90 and 95 per cent of the NRM was steeply inclined (Figs 7a, c), had coercivities between 0 and 20 mT (Figs 5, 7b, c), and unblocking temperatures between 20 and 400 °C (Fig. 6) The remaining 5–10 per cent of the NRM had an intermediate inclination and was much more resistant to both AF and thermal cleaning; it will be discussed in the next section.

The soft NRM vector is probably a drilling-induced remanence (DIRM; Burmester 1977), for the following reasons:

(1) The unblocking temperatures are too low for a total thermoremanence (TRM), too variable in their distribution for a partial TRM, and too high for a Brunhes-epoch viscous remanence (VRM).
(2) Q_n is several times higher than the expected value for TRM in MD magnetite (Stacey 1967) and much higher than the expected Q_n for VRM.
(3) The mean direction of the soft NRM is $D = 44°$, $I = +83°$ ($k = 37$, $\alpha_{95} = 3.4°$, $N = 48$ specimens) (see Fig. 8). The soft NRM is steeper than either the present geomagnetic field at the site ($D = 22°$, $I = +74°$) or the axial dipole field (the average field during the Brunhes: $D = 0°$, $I = +67°$). Part of the soft NRM may be a VRM, but much of it is a nearly vertical remanence.

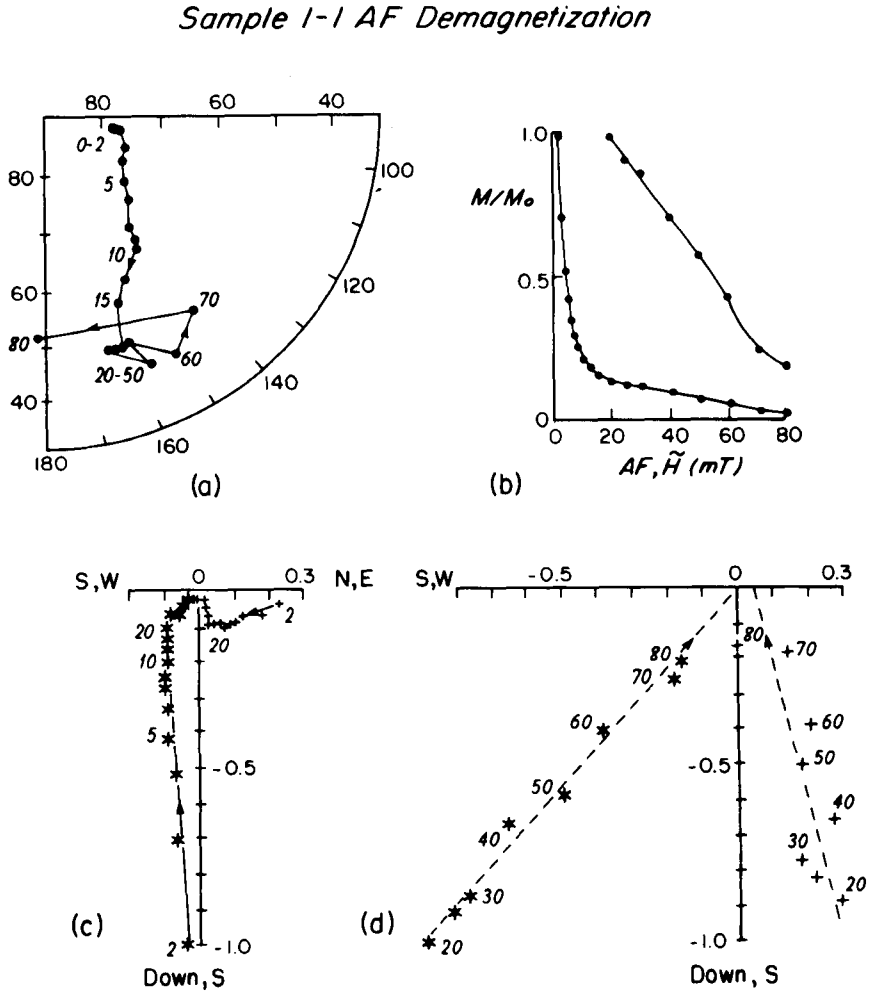


Figure 7. Typical changes in the direction and intensity of NRM during stepwise AF demagnetization of granodiorite samples, as illustrated by results for sample 1-1. (a) NRM directions at different levels of AF cleaning (labels are AFs in mT) plotted on part of a polar equal-area projection. All vectors are downward. (b) Total intensity decay curves of NRM. The upper curve is an enlargement of the lower curve above 20 mT and illustrates the coercivity spectrum characteristic of the hard NRM component. (c), (d) Orthogonal projections of NRM vectors in the course of AF cleaning on vertical (up vs. N, stars) and horizontal (N vs. E, crosses) planes. (d) is an enlargement of (c) above 20 mT, showing the univectorial character of the hard NRM component and its clean separation from the soft component.

Downloaded from https://academic.oup.com/gji/article/95/1/69/602275 by Curtin University Library user on 26 May 2026

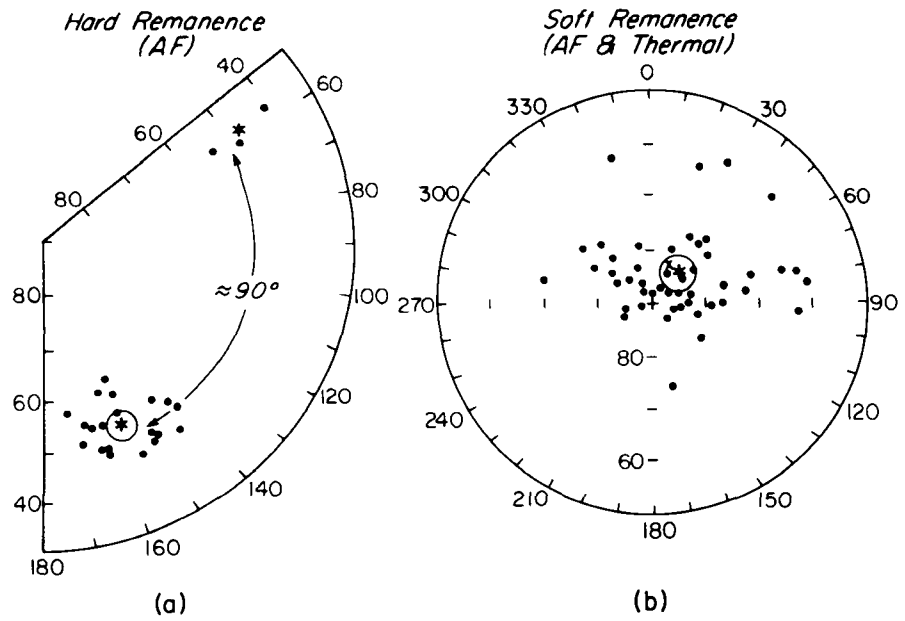


Figure 8. AF and thermally cleaned hard and soft remanence directions for individual specimens (dots), and their means (stars) and circles of 95 per cent confidence. The present Earth's field is indicated by a cross.

Steeply-inclined soft remanences due to drilling with steel core bits and barrels are familiar from DSDP and other borehole work (e.g. Ade-Hall & Johnson 1976). In granitic rocks, DIRM has been found to have coercivities primarily <15 mT, to affect coarse-grained rocks more than fine-grained ones, and to reside mainly in magnetic grains close to the cutting surface (Burmester 1977; Kodama 1984). In these respects, the soft NRM of our cores resembles DIRM (Figs 4, 5, 7). Jackson & Van der Voo (1985) concluded from a careful study of carbonate rocks that magnetites as fine as 1–2 μm in size acquire drilling and sawing remanences, which resist demagnetization to 60 mT or 400 °C. DIRM in our samples was erased by 20 mT or 400 °C cleaning and presumably resides in much coarser magnetites.

The mechanism of DIRM remains in doubt. Burmester (1977) found that DIRM had a $\log t$ time decay, in the manner of VRM, but only a small part of DIRM intensity could be accounted for by VRM acquired in the axial core-barrel field of ≈0.2 mT during a period of several weeks. Storage of our samples in the laboratory field of 60 μT for six weeks resulted in negligible VRM. For a drilling rate of ≈1 ft hr^{-1} in crystalline rock, our samples, which came from the deepest section cored, would have been exposed to the core-barrel field for only about one day.

Before dismissing VRM as a principal mechanism of DIRM, we should consider the possible role of elevated temperatures in amplifying VRM. Ambient temperatures of 60–70 °C are to be expected at the top of the basement in southern Alberta, and heating at the drillbit–rock interface could raise temperatures to perhaps 200 °C, even in the presence of drilling mud. Viscosity coefficients of submicron magnetites are about four times higher at 200 °C than at room temperature (Dunlop 1983b).

There is a suggestion in the thermal decay curves that edge specimens (1 and 4) do have higher average unblocking temperatures of the soft NRM component (Fig. 6) as well as higher intensities (Fig. 4). Figure 6 illustrates the extremes observed in thermal cleaning. Many edge specimens have decay curves like 20-1, although others have curves similar to 19-3, perhaps because the outermost heated material was largely trimmed from them in sectioning the transverse cores. Interior specimens have curves falling between the extremes represented by 19-3 and 2-3; their unblocking temperatures are lower on average than those of edge specimens.

There remains the problem that the spectrum of DIRM unblocking temperatures extends much above the 200 °C expected for even the highest-temperature short-term VRMs. Partial thermal remagnetization of the drillcore as it cooled to surface temperatures can be discounted for the same reason: partial TRMs could not have unblocking temperatures much above 200 °C.

Burmester (1977) and Kodama (1984) considered isothermal remanence (IRM) as a mechanism, but dismissed it because observed intensities and coercivities of DIRM required fields much greater than 60 μT. Axial fields well above the 0.2 mT measured by Burmester (1977) at the drill bit of a laboratory steel drill press could possibly exist at the cutting end of a long core barrel because of channelling of field lines through the steel barrel. The combination of enhanced fields and above-ambient temperatures might jointly produce high-temperature IRM/VRM with the requisite coercivities and unblocking temperatures.

Burmester (1977), Kodama (1984) and Jackson & Van der Voo (1985) all favoured a piezoremanent mechanism for DIRM. We tend to discount this explanation because the most highly stressed magnetite grains would be confined to the cutting surface, which was largely trimmed off our specimens (by sanding, not cutting). For the same reason, it is unlikely that contaminating steel particles in the drilling

Downloaded from https://academic.oup.com/gji/article/95/1/69/602275 by Curtin University Library user on 26 May 2026

mud (Halgedahl & Jarrard 1988) can be the root cause of DIRM. It is true that at a depth of 2500 m the entire drill core is subject to a lithostatic pressure of >0.5 kb, which will be released when the core is brought to the surface. Carmichael (1968) found that a magnetite rod acquired significant piezoremanence after release of a 0.2 kb stress, but the stress was uniaxial, not hydrostatic, and the fields applied (2–3 mT) were much larger than those likely in a core barrel.

A final possibility is that DIRM accounts for the lower unblocking temperatures in the soft NRM, and Brunhes-epoch VRM explains most of the higher unblocking temperatures. VRM unblocking temperatures up to 400 °C in magnetite are unusual, but not unknown (Kent 1985; Jackson & Van der Voo 1986). In this interpretation, the soft NRM would itself be a bivectorial combination of a steep VRM and a vertical DIRM, whose directions are not resolved in vector diagrams like Fig. 7(c). Vector swings from 0 to 2 mT AF cleaning often do have an easterly swing away from vertical at odds with the generally southerly vector swings at higher AFs (e.g. Fig. 7a), but many of the soft vectors lie far from the N–S vertical plane which would be expected to contain such composite vectors.

Whatever the origin of the soft NRM vector, it plays an important role in deducing the declination and even the polarity of stable NRM in unoriented cores (Ade-Hall & Johnson 1976; Kodama, 1984). From Fig. 8(b), it is obvious that in the present study declinations of individual soft NRMs are widely scattered. They would be a poor guide to present geographic or geomagnetic north for individual specimens; the multishot orientation technique is more trustworthy in the case of our core. Even the *average* declination of 44° is 22° east of geomagnetic north and would lead to a significant error if used in pole determination.

6 CHARACTERISTIC NRM

A harder NRM vector with SSE declination and intermediate positive inclination was isolated in 20 of the 23 granodiorite samples. Although this NRM forms only 5–10 per cent of the total, it was cleanly resolved as an endpoint in stereoplots (Fig. 7a) and by principal component analysis (Kirschvink 1980) of linear segments in vector diagrams (Fig. 7d). The inclinations of the hard magnetization in the three border specimens included in the averaging were 47.5°, 59° and 62° (1.1, 2.4 and 8.1, Table 1). These values are similar to the 45°–60° range of inclinations for the 17 interior specimens. There is no suggestion of residual contamination by undemagnetized DIRM in the border specimens which might bias the average hard NRM direction.

The characteristic NRM has coercivities in the range 20–80 mT, and the shape of its coercivity spectrum is diagnostic of single-domain (SD) magnetite (Dunlop 1983a) (Figs 7b, d). It has discrete unblocking temperatures concentrated just below 580 °C.

This fine SD-like magnetite is distinct from the coarse, secondary magnetite that carries drilling remanence. It may occur as primary inclusions in feldspars. The samples are relatively fresh and unaltered, as noted earlier. The large batholith from which they come has undergone no higher than greenschist-facies regional metamorphism, implying reheating to no more than 500 °C. Since the SD magnetite has NRM unblocking temperatures close to 580 °C, the stable NRM is probably a primary TRM, rather than a thermoviscous overprint. The pluton may be a late-stage intrusion, which escaped the full force of regional metamorphism and accompanying magnetic overprinting.

Three adjacent granodiorite specimens (21.1, 22.2, 23.4) from the 2562 m level have similar inclinations of hard NRM to other cores but declinations that are rotated approximately 90° (Fig. 8a). This section of core appears to have been rotated accidentally before marking and transverse drilling. Another possibility is that this section broke and rotated relative to the rest of the core before it was scribed (cf. Bleakly *et al.* 1985, Fig. 8). The two granite samples from the 2560 m level did not yield well-defined stable NRMs.

A depth log of NRM inclination, MDF and intensity is given in Fig. 9. The intensities of hard and soft NRMs have very similar depth variations. At all horizons sampled except 2563 m, the hard NRM averages 7–8 per cent of the soft NRM. The hard and soft NRMs were acquired by quite different physical processes, but they reside in different fractions of a single magnetic phase with a relatively constant grain-size distribution.

The inclination of the stable NRM is relatively constant between +50° and +55° in the top 5–6 m of the core, then decreases to about +45° in the bottom 1–2 m. NRM intensity also changes significantly at the base of the core. These differences may reflect paleosecular variation (PSV) during slow cooling of the pluton, although the dispersion of I values at any one level is similar to the total variation in average I down the core. In order to average out PSV, the uppermost 7 m of the pluton would have had to cool over about 10 000 yr. Since we do not know the size of the pluton, it is safer to assume that PSV has not been averaged out.

The hard NRM direction after AF cleaning is $D = 157°$, $I = +52°$ ($k = 118$, $\alpha_{95} = 3°$, $N = 20$). Thermal cleaning did not separate the hard and soft NRMs as cleanly; these data were not averaged.

7 VGP AND AGE OF MAGNETIZATION

A sketch of the Laurentian apparent polar wander path (APWP) between 2200 and 1650 Ma appears in Fig. 10. The exact extent of the 1900–1700 Ma path (the Coronation Loop of Irving & McGlynn 1979) is uncertain, but its general trend is well established. The main group of 20 cleaned hard NRMs give a VGP at 4°S, 93°W ($dp = 3°$, $dm = 4°$), falling on the APWP around 1850 Ma. Even allowing for PSV, the age of magnetization is likely between 1900 and 1800 Ma.

Uplift and cooling overprint magnetizations from the Churchill, Superior and North Atlantic cratons tend to fall on the ascending limb of the Coronation Loop, between 1750 and 1650 Ma. Late-stage Hudsonian K/Ar and Rb/Sr ages are also in this general range (e.g. Green *et al.* 1985, Table 1). The position of the PCP Travers pole on the older descending limb of the Coronation Loop implies magnetization earlier during Hudsonian events. Most likely the parent pluton formed around 1850 Ma and the hard NRM is a primary TRM, as we concluded earlier on the basis of its

Downloaded from https://academic.oup.com/gji/article/95/1/69/602275 by Curtin University Library user on 26 May 2026

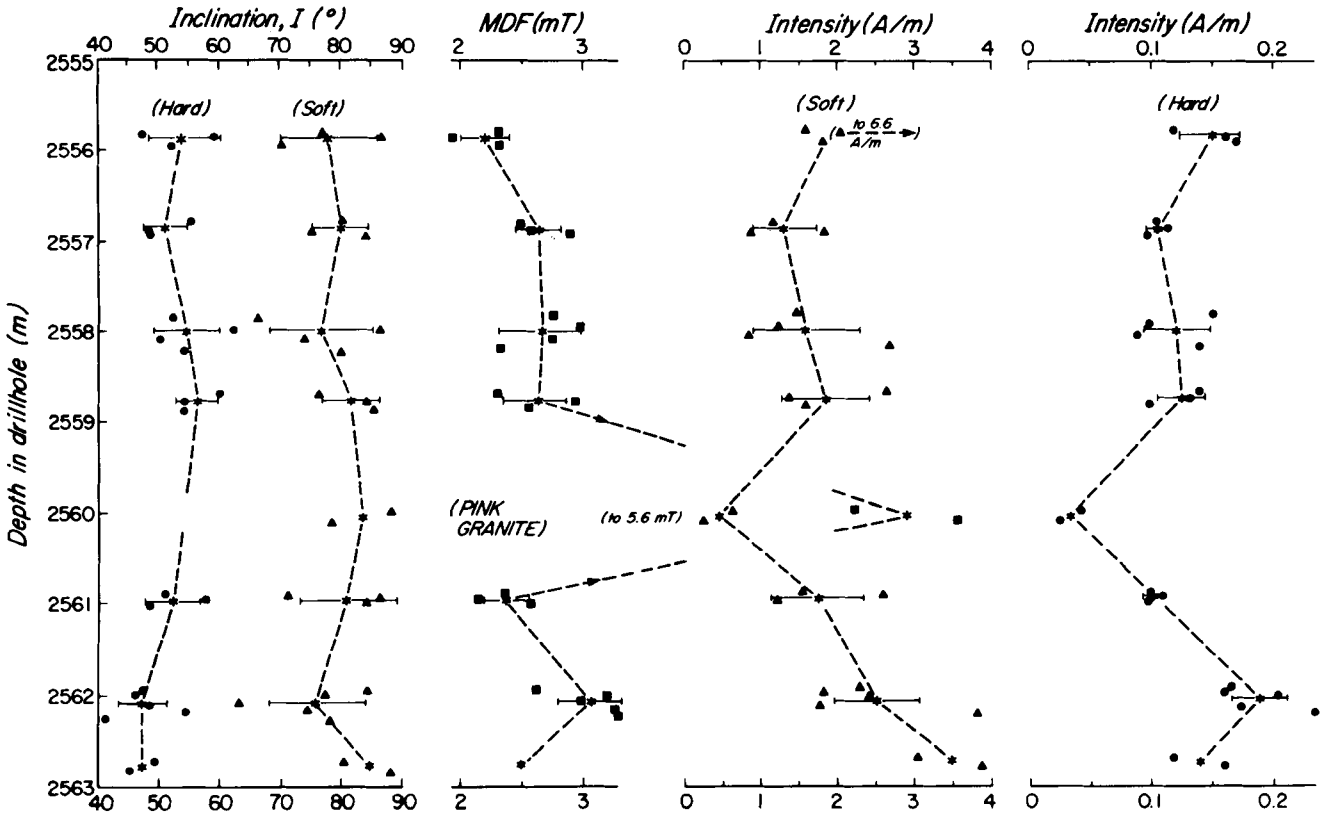


Figure 9. A depth log of inclinations and intensities of hard (triangles) and soft (circles) NRMs and the MDF of the composite remanence (squares).

high unblocking temperatures. It is also possible that the pluton is older than 1850 Ma but was magnetically overprinted at this time.

8 DISCUSSION AND CONCLUSIONS

There is a certain amount of controversy about the extensions of some domain boundaries in the western Churchill Province to the south and west under Phanerozoic cover (Van Schmus & Bickford 1981; Thomas & Gibb 1985; Green *et al.* 1985, 1986; Lewry *et al.* 1986). In the new tectonic sub-division suggested by Hoffman (1987), the basement at the PCP Travers drill-site lies in the South Keewatin Platform of the western Churchill, just north of the Archean Wyoming craton and well to the west of the Trans-Hudson Orogen.

The magnetization of the PCP Travers core certainly dates from the time of Hudsonian reactivation in the Churchill. However, the NRM does not seem to be a secondary thermal or thermochemical overprint. The fresh condition and relatively low metamorphic grade of the rocks (Boyle 1985) together with the high NRM unblocking temperatures suggest magnetization during initial cooling, at some distance from a collision and suture zone. In fact, 1850 Ma is quite a commonly observed age of island-arc and other magmatism in the western Churchill (U/Pb zircon dates: Green *et al.* 1985; Table 1; Van Schmus *et al.* 1987, Table 1). The PCP Travers pluton likely formed during this magmatic episode.

We draw the following general conclusions from this study.

(1) It is possible to obtain high-quality paleomagnetic data from oriented deep continental drillcores. This has only rarely been done before (e.g. Michigan Basin borehole, Van der Voo & Watts 1978; Alaska North Slope drillcores, Halgedahl & Jarrard 1988).
(2) VGPs from drillcores can be used to date the basement quite precisely if PSV was low or if cooling following regional metamorphism was sufficiently slow that PSV averages out over the section sampled. These magnetic basement ages are a valuable supplement to conventional radiometric dates.
(3) Felsic to intermediate plutonic rocks containing mainly soft MD magnetite in some cases retain a stable primary or synorogenic NRM that can be cleanly resolved from strong secondary NRMs.

ACKNOWLEDGMENTS

It is a pleasure to acknowledge the help of Peter Savage of PanCanadian Petroleum Ltd., who suggested the project and supplied the core samples and analyses. Don McKanlew of Norton Christensen Canada Ltd and Susan Halgedahl of Lamont–Doherty Geological Observatory gave useful information on the core orienting system and log. Ken Kodama, an anonymous referee and members of the Erindale brown-bag seminar group made numerous

Downloaded from https://academic.oup.com/gji/article/95/1/69/602275 by Curtin University Library user on 26 May 2026

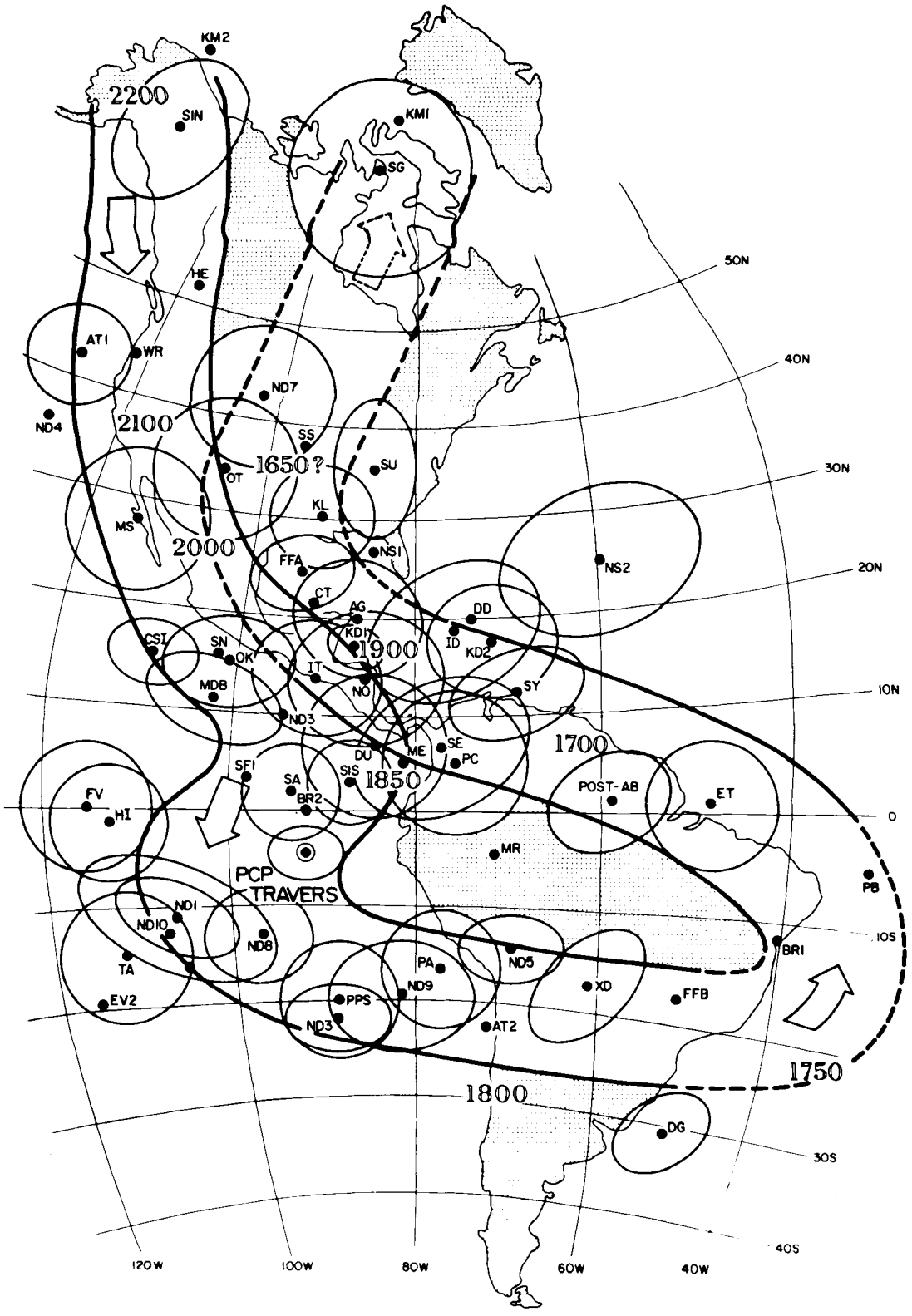


Figure 10. APWP for Laurentia during the interval 2200–1650 Ma. Palaeopole labels follow Irving (1979) and Irving & McGlynn (1981), with the following additions: CSI, Cape Smith igneous; FV, Flaherty volcanics; HI, Haig intrusions; EV2, Eskimo volcanics overprint: all from Schmidt (1980); POST-AB, post-Abitibi overprint: Schutts & Dunlop (1981); OK, Ottawa Islands komatiites: Buchan & Baragar (1985). The VGP for the PCP Travers core falls on the APWP around 1850 Ma.

Downloaded from https://academic.oup.com/gji/article/95/1/69/602275 by Curtin University Library user on 26 May 2026

suggestions and criticisms that improved the final paper. This research was supported by Natural Sciences and Engineering Research Council of Canada Operating Grant A7709 to DJD.

REFERENCES

Ade-Hall, J. M. & Johnson, H. P., 1976. Paleomagnetism of basalts, Leg 34, *Init. Rep. deep Sea Drill. Proj.*, **34,** 513–532.

Bleakly, D. C., Van Alstine, D. R. & Packer, D. R., 1985. Core orientation 1. Controlling errors minimizes risk and cost in core orientation. 2. How to evaluate orientation data, quality control, *Oil Gas J.*, 1985 Dec. 2 103–109 and 1985 Dec. 9, 46–52.

Boyle, P. J., 1985. *Description and interpretation of a special granite core 2550.0 m–2563.2 m (Southern Alberta), PanCanadian Petroleum Internal Report, PCP Travers 1-31-13-20W4.*

Buchan, K. L. & Baragar, W. R. A., 1985. Paleomagnetism of the komatitic basalts of the Ottawa Islands, N. W. T., *Can. J. Earth Sci.*, **22,** 553–556.

Burmester, R. F., 1977. Origin and stability of drilling induced remanence, *Geophys. J. R. astr. Soc.*, **48,** 1–14.

Carmichael, R. S., 1968. Remanent and transitory effects of elastic deformation of magnetite rods, *Phil. Mag.*, **17,** 911–927.

Dunlop, D. J., 1983a. Determination of domain structure in igneous rocks by alternatig field and other methods, *Earth planet. Sci. Lett.*, **63,** 353–367.

Dunlop, D. J., 1983b. Viscous magnetization of 0.04–100 μm magnetites, *Geophys. J. R. astr. Soc.*, **74,** 667–687.

Evans, M. E. & Maillol, J. M., 1986. A palaeomagnetic investigation of a Permian redbed sequence from a mining drill core, *Geophys. J. R. astr. Soc.*, **87,** 411–419.

Green, A. G., Hajnal, Z. & Weber, W., 1985. An evolutionary model of the western Chruchill Province and western margin of the Superior Province in Canada and the north-central United States, *Tectonophys.*, **116,** 281–322.

Green, A. G., Hajnal, Z., & Weber, W., 1986. An evolutionary model of the western Churchill Province and western margin of the Superior Province in Canada and the north-central United States—reply, *Tectonophys.*, **131,** 188–197.

Halgedahl, S. L. & Jarrard, R. D., 1988. Paleomagnetism of the Kuparuk River formation from oriented drill core: evidence for rotation of the Arctic Alaska Plate, in *Alaskan North Slope Geology,* Pacific Soc. Econ. Palaeont. Mineral., in press.

Heider, F., Dunlop, D. J. & Sugiura, N., 1987. Magnetic properties of hydrothermally recrystallized magnetite crystals, *Science,* **236,** 1287–1290.

Hoffman, P. F., 1987. Birth of a craton (abstract), *Geol. Ass. Can. Progr. with Abstr.*, **12,** 56.

Irving, E., 1979. Paleopoles and paleolatitudes of North America and speculations about displaced terrains, *Can. J. Earth Sci.*, **16,** 669–694.

Irving, E. & McGlynn, J. C., 1979. Palaeomagnetism in the Coronation Geosyncline and arrangement of the continents in the middle Proterozoic, *Geophys. J. R. astr. Soc.*, **58,** 309–336.

Irving, E. & McGlynn, J. C., 1981. On the coherence, rotation and palaeolatitude of Laurentia in the Proterozoic, in *Precambrian Plate Tectonics,* pp. 561–598, ed. Kröner, A., Elsevier, Amsterdam.

Jackson, M. & Van der Voo, R., 1985. Drilling-induced remanence in carbonate rocks: occurrence, stability and grain-size dependence, *Geophys. J. R. astr. Soc.*, **81,** 75–87.

Jackson, M. & Van der Voo, R., 1986. Thermally activated viscous remanence in some magnetite- and hematite-bearing dolomites, *Geophys. Res. Lett.*, **13,** 1434–1437.

Kent, D. V., 1985. Thermoviscous remagnetization in some Appalachian limestones, *Geophys. Res. Lett.*, **12,** 805–808.

Kirschvink, J. L., 1980. The least-squares line and plane and the analysis of paleomagnetic data, *Geophys. J. R. astr. Soc.*, **62,** 699–718.

Kodama, K. P., 1984. Palaeomagnetism of granitic intrusives from the Precambrian basement under eastern Kansas: orienting drill cores using secondary magnetization components, *Geophys. J. R. astr. Soc.*, **76,** 273–287.

Lewry, J. F., Collerson, K. D., Bickford, M. E. & Van Schmus, W. R., 1986. An evolutionary model of the western Churchill Province and western margin of the Superior Province in Canada and the north-central United States—discussion, *Tectonophys.*, **131,** 183–188.

Nelson, R. A., Lenox, L. C. & Ward, B. J. Jr., 1987. Oriented core: its use, error, and uncertainty, *Bull. Am. Ass. Petrol. Geol.* **71,** 357–367.

Rowley, D. S., Burk, C. A. & Manuel, T., 1981. *Oriented Cores,* Norton-Christensen Technical Report.

Schmidt, P. W., 1980. Paleomagnetism of igneous rocks from the Belcher Islands, Northwest Territories, Canada, *Can. J. Earth Sci.*, **17,** 807–822.

Schutts, L. D. & Dunlop, D. J., 1981. Proterozoic magnetic overprinting of Archaean rocks in the Canadian Superior Province, *Nature,* **291,** 642–645.

Stacey, F. D., 1967. The Koenigsberger ratio and the nature of thermoremanence in igneous rocks, *Earth planet. Sci. Lett.*, **2,** 67–68.

Thomas, M. D. & Gibb, R. A., 1985. Proterozoic plate subduction and collision: processes for reactivation of Archean crust in the Churchill Province, *Spec. Pap. geol. Ass. Can.*, **28,** 263–279.

Van der Voo, R. & Watts, D. R., 1978 Paleomagnetic results from igneous and sedimentary rocks from the Michigan Basin borehole, *J. geophys. Res.*, **83,** 5844–5848.

Van Schmus, W. R. & Bickford, M. E., 1981. Proterozoic chronology and evolution of the midcontinent region, North America, in *Precambrian Plate Tectonics,* pp. 261–296, ed. Kröner, A., Elsevier, Amsterdam.

Van Schmus, W. R., Bickford, M. E., Lewry, J. F. & Macdonald, R., 1987. U-Pb geochronology in the Trans-Hudson Orogen, northern Saskatchewan, Canada, *Can. J. Earth Sci.*, **24,** 407–424.

Vine, F. J. & Matthews, D. H., 1963. Magnetic anomalies over ocean ridges, *Nature,* **199,** 947–949.

Downloaded from https://academic.oup.com/gji/article/95/1/69/602275 by Curtin University Library user on 26 May 2026